

Anaerobic Co-digestion of Food Waste and Glycerol in Single-Stage System

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Abstract

The growing generation of food waste and the surplus of crude glycerol from biodiesel production represent two significant organic waste streams requiring sustainable management solutions. This study evaluated the technical feasibility of a single-stage anaerobic co-digestion system treating food waste and glycerol as co-substrates, with the objective of verifying biogas production while providing a valorisation route for crude glycerol. The investigation was conducted in a 1 L continuous stirred-tank reactor operated under mesophilic conditions (35 °C) with a hydraulic retention time of 30 days. Food waste was collected from a university canteen, and crude glycerol was obtained from biodiesel production residues; anaerobic sludge from a municipal wastewater treatment plant served as inoculum. The reactor was operated in sequential phases: an initial adaptation period, co-digestion of food waste and glycerol, glycerol only feeding, and a post-feeding period. Biogas production was monitored continuously using an automatic methane potential test system, while process stability was assessed through measurements of pH, total alkalinity, volatile fatty acids, and their ratio. Substrate characterisation included total solids, volatile solids, density, total carbon, and total nitrogen analyses. Results demonstrated that co-digestion of food waste and glycerol yielded the highest methane production rates and cumulative volumes, with flow spikes immediately following feeding events indicating rapid substrate conversion. In contrast, glycerol-only feeding resulted in reduced biogas production despite continued methanogenic activity. Total alkalinity remained elevated following bicarbonate supplementation, maintaining pH around 7 throughout most of the experiment; volatile fatty acid concentrations fluctuated between 3000 to 5000 mg CH₃COOH L⁻¹, with *VFA/alkalinity* ratios of 0.3 to 1.2. These findings confirm the hypothesis that glycerol can be useful for biogas production when co-digested with food waste or alone. The study demonstrates the technical viability of anaerobic co-digestion as a strategy for valorising crude glycerol within a circular bioeconomy framework, though further research is needed to optimise glycerol loading rates and develop robust feeding strategies for industrial application.

Resumo

A crescente geração de resíduos alimentares e o excedente de glicerol bruto proveniente da produção de biodiesel representam duas produções significativas de resíduos orgânicos que requerem soluções sustentáveis. Este estudo avaliou a viabilidade técnica de um sistema de codigestão anaeróbia de uma única etapa, tratando resíduos alimentares e glicerol como co-substratos, com o objetivo de verificar a produção de biogás e, simultaneamente, proporcionar uma via de valorização para o glicerol bruto. A investigação foi conduzida num reator contínuo de tanque agitado de 1 L, operado em condições mesofílicas (35 °C), com um tempo de retenção hidráulica de 30 dias. Os resíduos alimentares foram recolhidos numa cantina universitária e o glicerol bruto foi obtido a partir de resíduos da produção de biodiesel; lamas anaeróbias de uma estação de tratamento de águas residuais municipais serviram como inóculo. O reator foi operado em fases sequenciais: um período inicial de adaptação, codigestão de resíduos alimentares e glicerol, alimentação exclusiva com glicerol e um período pós-alimentação. A produção de biogás foi monitorizada continuamente através de um *automatic methane potential test system*, enquanto a estabilidade do processo foi avaliada por medições de pH, alcalinidade total, ácidos graxos voláteis e respectiva razão. A caracterização dos substratos incluiu análises de sólidos totais, sólidos voláteis, densidade, carbono total e nitrogênio total. Os resultados demonstraram que a codigestão de resíduos alimentares e glicerol originou as taxas de produção de metano e os volumes acumulados mais elevados, com picos imediatamente após as alimentações, indicando uma rápida conversão do substrato. Em contraste, a alimentação exclusiva com glicerol resultou numa produção reduzida de biogás, apesar da atividade metanogénica continuada. A alcalinidade total manteve-se elevada após a suplementação com bicarbonato, mantendo o pH em torno de 7 durante a maior parte da experiência; as concentrações de ácidos graxos voláteis flutuaram entre 3.000 a 5.000 mg CH₃COOH L⁻¹, com razões *AGV/alcalinidade* de 0,3 a 1,2. Estes resultados confirmam a hipótese de que o glicerol pode ser útil para a produção de biogás quando codigerido com resíduos alimentares ou isoladamente. O estudo demonstra a viabilidade técnica da codigestão anaeróbia como estratégia de valorização do glicerol bruto no âmbito de uma bioeconomia circular, embora seja necessária investigação adicional para otimizar as cargas de glicerol e desenvolver estratégias de alimentação

robustas para aplicação industrial.

Contents

1	Introduction	1
2	Objectives	3
2.1	General Objectives	3
2.2	Specific Objectives	3
3	Literature Review	4
4	Materials and Methods	8
4.1	Collection and Preparation of the Food Waste	8
4.2	Crude Glycerol Collection	8
4.3	Inoculum Collection	8
4.4	Physical and Chemical Characterisation	9
4.4.1	TS and VS Determination	9
4.4.2	Density Determination	11
4.4.3	TOC and TN Determination	11
4.5	Experiment Configuration	12
4.5.1	Feeding Solution	12
4.5.2	Biogas Quantification	13
4.5.3	Analysis Methods	13
4.5.4	Substrate used as feed	15
5	Results	17
5.1	Reactor Condition and Visual Observations	17
5.2	Biogas Production	17
5.2.1	Overall Production	19
5.2.2	Adaptation Phase	21
5.2.3	Co-digestion of Food Waste and Crude Glycerol	22
5.2.4	Digestion of Crude Glycerol	24
5.2.5	Post-feeding Period	26

5.2.6 Alkalinity and Volatile Fatty Acids	27
6 Conclusion	30
Abbreviations	31
Symbols	32
Units	33
References	34

Chapter 1

Introduction

The generation of organic waste has become one of the defining environmental challenges of our time, particularly as urbanization and industrial activity continue to expand. A major contributor is food waste, a global stream that in the European Union alone is estimated at 88 Mt yr⁻¹, excluding inedible parts removed from the supply chain (Scherhauser et al., 2018). When considering only post-processing activities – wholesale, retail, food service and households – approximately 66.3 Mt of food waste are produced annually in the EU-27+1 (Albizzati et al., 2021). The scale of this loss highlights the urgent need for improved management strategies that move beyond simple disposal.

The environmental burden of food waste is substantial. It accounts for roughly 15 % to 16 % of the total climate impact of the entire food supply chain, translating to about 186 Mt CO₂-eq of greenhouse gas emissions, alongside significant contributions to acidification and eutrophication (Scherhauser et al., 2018). The majority of these impacts arise during primary production, making prevention the single most effective strategy; however, when waste cannot be avoided, its valorisation through anaerobic digestion or composting is favoured over landfill disposal (Albizzati et al., 2021; Scherhauser et al., 2018). Despite this, much food waste continues to be landfilled or incinerated, forfeiting the opportunity to recover its embedded energy and nutrients.

Parallel to the food waste challenge, the rapid expansion of biodiesel production has generated an excess of another organic by-product: crude glycerol. Formed in the transesterification of triglycerides, approximately 10 kg of crude glycerol are produced for every 100 kg of biodiesel (Garlapati et al., 2016). Global biodiesel output, driven by renewable energy mandates, has caused the glycerol market to swell – with crude glycerol supply projected to reach about 6.33 Mt by 2025, of which nearly 680 kt are expected to be highly impure due to the shift towards waste-based feedstocks (Attarbachhi et al., 2023).

Crude glycerol composition varies widely depending on the oil source and catalyst used, but it typically contains methanol, soaps, free fatty acids, salts and other matter organic

non-glycerol (MONG) compounds (Anand & Saxena, 2012; Attarbachhi et al., 2023). These impurities are known to inhibit microbial growth and fermentation, as demonstrated by the severe reduction in 1,3-propanediol production when untreated crude glycerol from jatropha, soybean or sunflower oils was used (Anand & Saxena, 2012). Purification to technical or pharmaceutical grade is energy-intensive and often uneconomical for small and medium biodiesel plants (Attarbachhi et al., 2023), prompting the search for direct, cost-effective valorisation routes. Because of its high organic content and ready biodegradability, crude glycerol has attracted attention as an ideal co-substrate for anaerobic digestion, where it can serve as a concentrated carbon source while being simultaneously detoxified by dilution and microbial synergy (Gebreegziabher et al., 2025; Tomczak et al., 2025).

AD is a mature biological process in which consortia of microorganisms convert organic matter into biogas – primarily methane (55 % to 65 %) and carbon dioxide – under oxygen-free conditions, passing through the sequential stages of hydrolysis, acidogenesis, acetogenesis and methanogenesis (Gebreegziabher et al., 2025). AD offers a unique combination of benefits: it produces renewable, storable energy that can replace fossil fuels; generates a nutrient-rich digestate that can be used as organic fertiliser; and reduces the environmental load of organic wastes by avoiding landfilling and its associated methane emissions (Gebreegziabher et al., 2025; Scherhauser et al., 2018).

Within this framework, anaerobic co-digestion – the simultaneous treatment of two or more substrates – has emerged as a particularly effective strategy. Co-digesting crude glycerol with nitrogen-rich materials such as animal manure, sewage sludge or food waste balances the carbon-to-nitrogen ratio, dilutes potential inhibitors, provides essential macro- and micronutrients, and fosters synergistic microbial interactions (Gebreegziabher et al., 2025; Tomczak et al., 2025). The result can be a dramatic improvement in biogas and methane yield: studies report increases ranging from 14 % to 226 % when crude glycerol is added to a variety of base substrates, with methane contents generally exceeding 60 % as long as glycerol does not exceed about 8 % (v/v) of the feed (Gebreegziabher et al., 2025; Tomczak et al., 2025).

However, excessive glycerol loads can lead to volatile fatty acid accumulation, pH drop and methanogen inhibition, underscoring the need for careful optimisation (Gebreegziabher et al., 2025; Tomczak et al., 2025). Overall, the convergence of abundant food waste and surplus crude glycerol supplies, together with the proven capability of anaerobic digestion to convert these streams into renewable methane and fertiliser, represents a tangible route towards a circular, low-carbon bioeconomy.

Chapter 2

Objectives

2.1 General Objectives

The primary objective of this dissertation project is to evaluate the efficiency of a single-stage anaerobic co-digestion system using organic food waste and glycerol as substrates. The investigation will be conducted by determining biogas production, with a focus on industrial feasibility.

The central hypothesis is that the controlled addition of glycerol, a by-product of the biodiesel industry, enhances biogas production due to its high energy content and rapid biodegradability, provided that adequate operational conditions are maintained to avoid volatile fatty acid overload.

2.2 Specific Objectives

1. Characterize the substrates (food waste and glycerol), evaluating physicochemical parameters relevant to anaerobic digestion.
2. Operate the single-stage anaerobic co-digestion system, assessing process stability.
3. Evaluate the techno-economic feasibility of using glycerol as a co-substrate, considering energy yield and potential for industrial application.

Chapter 3

Literature Review

The rapid expansion of biodiesel production has generated an oversupply of crude glycerol, a by-product of the transesterification output, this fig. 3.1 shows the steps to produce biodiesel and when glycerol is produced (Garlapati et al., 2016). This waste stream, loaded with impurities such as methanol, soaps, and hydroxides, the fig. 3.2 shows the processes and what they contaminate the glycerol with, poses environmental disposal challenges, also with the variation of impurities depending on the source, there's no standard method to use this source of carbon (Anand & Saxena, 2012; Gebreegziabher et al., 2025). Anaerobic digestion (AD) aligns with circular economy principles by converting organic residues into biogas while recycling nutrients, making crude glycerol a promising co-substrate (Gebreegziabher et al., 2025).

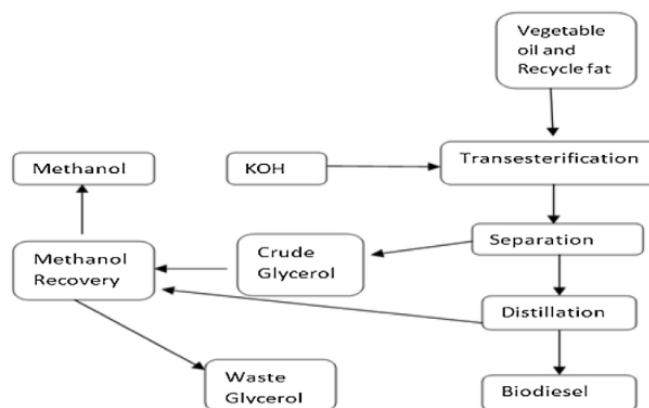


Figure 3.1: Source: Garlapati et al. (2016)

Gebreegziabher et al. (2025) reported that co-digestion of crude glycerol can boost methane production compared to mono-digestion of the base substrate. Tomczak et al. (2025) reviewed one-stage AD experiments and concluded that a methane content above 60 % could be consistently obtained when the crude glycerol fraction did not exceed 8 % (v/v) of the feed, the flow chart by Li et al. (2011) on the fig. 3.3 it shows that using AD

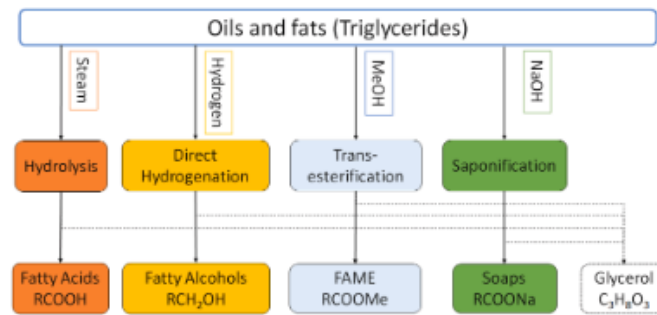


Figure 3.2: Source: Attarbach et al. (2023)

there could be up to 70 % methane production. However, the same review emphasized that increasing the glycerol loading beyond a system-specific threshold may provoke inhibitory effects, likely due to the accumulation of volatile fatty acids (VFAs) and the impurities present in crude glycerol (Anand & Saxena, 2012; Tomczak et al., 2025). The heterogeneous composition of crude glycerol, affected by oil feedstock, catalyst, and biodiesel processing, demands careful feedstock characterization and, if necessary, simple pretreatment to avoid process upsets (Attarbach et al., 2023).

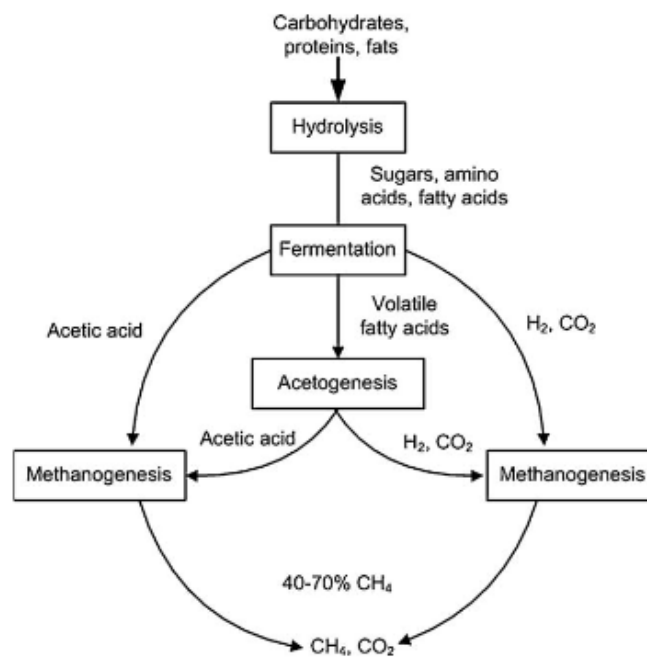
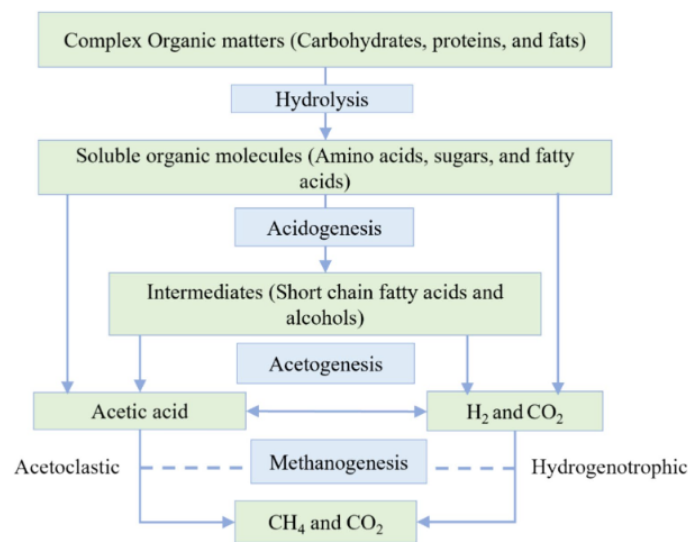


Figure 3.3: Source: Li et al. (2011)

Liquid anaerobic digestion (AD), typically conducted in a continuous stirred tank reactor (CSTR), operates at total solid concentrations of 0.5 % to 15 % and is generally used for treating sewage sludge, manure, and food waste (Li et al., 2011). The CSTR's continuous mixing provides uniform distribution of substrates, microorganisms, and heat, which is

essential when feeding a fermentable substrate. Temperature is a critical operational parameter; the activity of methanogenic consortia is strongly temperature dependent, and both mesophilic and thermophilic regimes have been applied to AD (Choorit & Wisarnwan, 2007). Tomczak et al. (2025) noted that the majority of glycerol co-digestion studies have been performed under mesophilic conditions, while thermophilic operation, which could offer higher conversion rates, still requires more systematic investigation. The CSTR platform is well suited to explore such temperature effects because its design allows easy control of heating and hydraulic retention time.

The selection of a robust microbial inoculum is decisive for rapid start-up and stable performance of a CSTR fed with glycerol. Anaerobic sludge from municipal wastewater treatment plants (WWTPs) is the most common source of inoculum because it harbours a diverse consortium of hydrolytic, acidogenic, and methanogenic microorganisms, the flow chart from Gebreegziabher et al. (2025) on the fig. 3.4 shows each step of the AD until the methane production. Posmanik et al. (2020) demonstrated that granular sludge, obtained from up-flow anaerobic sludge blanket reactors treating wastewater, outperforms suspended biomass as an inoculum. Although their work focused on batch tests, the findings strongly suggest that using granular sludge from a WWTP as seed for a CSTR could accelerate acclimation, improve process stability, and reduce the risk of acidification during load increases.



the importance of accurate VFA and alkalinity measurements for timely detection of process imbalance. Using the method from DiLallo & Albertson (Ribas et al., 2007) provide a validated method for measuring VFAs and alkalinity. Maintaining a favourable VFA-to-alkalinity ratio is particularly critical when co-digesting glycerol, as excess VFAs can inhibit methanogenesis (Tomeczak et al., 2025).

The available literature confirms that crude glycerol is a high-energy substrate that can significantly boost methane production in liquid AD systems. A CSTR provides the mixing and temperature control required to harness this potential. Using granular sludge from a wastewater treatment plant as the inoculum can enhance start-up speed and methanogenic activity, as demonstrated by comparative studies. To unlock the full benefits while avoiding inhibition, the glycerol loading must be kept below documented thresholds, and the reactor state should be continuously tracked through standard measurements of VFAs and alkalinity. Further research is needed to optimise thermophilic CSTR operation and to establish long-term performance data for granular-sludge-inoculated reactors treating crude glycerol.

Chapter 4

Materials and Methods

4.1 Collection and Preparation of the Food Waste

Food waste was collected from the canteen at the Instituto Politécnico de Bragança (IPB), prioritizing residues with high proportions of rice, pasta, and bread to maximize the sugar content and minimize the nitrogen content. The waste was diluted with approximately 2.5 L of distilled water and homogenised using a blender to obtain a liquid paste. It was then stored in a refrigerator at 4 °C to prevent spoilage and suppress microbial and fungal activity (Degueurce et al., 2020).

4.2 Crude Glycerol Collection

A 6 L bottle of crude glycerol, obtained from previous biodiesel production experiments at IPB, was provided for this study. Crude glycerol from biodiesel typically contains impurities such as catalyst residues (Attarbachí et al., 2023), which may influence its behaviour in anaerobic digestion.

4.3 Inoculum Collection

The anaerobic inoculum consisted of sludge collected from Águas do Norte wastewater treatment plant in Bragança. This ensured the availability of a microbial population for the anaerobic digestion (Posmanik et al., 2020).

4.4 Physical and Chemical Characterisation

The total solids (TS), volatile solids (VS), fixed solids (FS), and density were determined for the food waste, crude glycerol, and inoculum. Total organic carbon (TOC) and total nitrogen (TN) were determined for the food waste and crude glycerol.

4.4.1 TS and VS Determination

TS and VS were measured using a sequential gravimetric method based on Standard Methods 2540 B and 2540 E (Baird et al., 2017). The method involves controlled drying and calcination steps to quantify total solids and volatile solids fractions.

Initially, porcelain crucibles were placed in a muffle furnace at 550 °C for 1 h. For cooling, they were transferred to an oven at 105 °C for 15 min and then moved to a desiccator until they reached room temperature; their empty masses were recorded. For each material, a sample was placed into three crucibles and the wet mass was measured. The crucibles were then dried in an oven at 105 °C for 2 h (crude glycerol required 6 h to 8 h to reach constant mass). After drying, the crucibles were cooled in a desiccator to room temperature and weighed again to determine TS.

The same crucibles were subsequently placed in a muffle furnace at 550 °C for 50 min. Afterwards they were transferred to an oven at 105 °C for 15 min and then moved to a desiccator until they reached room temperature; the final masses were recorded to calculate VS.

TS, VS and FS were calculated as follows:

$$TS = \frac{M_{105} - M_{crucible}}{V_{sample}} \quad [g L^{-1}] \quad (4.1)$$

$$VS = \frac{M_{105} - M_{550}}{V_{sample}} \quad [g L^{-1}] \quad (4.2)$$

$$FS = \frac{M_{550} - M_{crucible}}{V_{sample}} \quad [g L^{-1}] \quad (4.3)$$

where M_{wet} is the mass of crucible plus wet sample, M_{105} is the mass after drying at 105 °C, M_{550} is the mass after ignition at 550 °C, $M_{crucible}$ is the mass of the empty crucible and V_{sample} is the volume of the sample. The volume is in L and the weight is in g. The results are on the table 4.1.

Table 4.1: Total Solids, Volatile Solids and Fixed Solids values

Sample		M_{crucible} (g)	M_{wet} (g)	M_{sample} (g)	V_{sample} (L)	M_{105} (g)	TS (g L ⁻¹)	Average TS (g L ⁻¹)	M_{550} (g)	VS (g L ⁻¹)	Average VS (g L ⁻¹)	FS (g L ⁻¹)	Average FS (g L ⁻¹)
Food Waste	1.000	7.873e1	1.060e2	2.726e1	2.550e-2	9.100e1	4.810e2		7.920e1	4.625e2		1.848e1	
	2.000	8.432e1	1.133e2	2.898e1	2.710e-2	9.887e1	5.364e2	5.364e2	8.474e1	5.208e2	5.208e2	1.559e1	1.848e1
	3.000	7.759e1	1.088e2	3.125e1	2.920e-2	9.344e1	5.420e2		7.818e1	5.216e2		2.038e1	
Crude Glycerol	1.000	8.422e1	8.864e1	4.424	5.000e-3	8.850e1	8.634e2		8.421e1	8.644e2		-9.891e-1	
	2.000	7.348e1	7.799e1	4.509	5.000e-3	7.786e1	8.677e2	8.677e2	7.347e1	8.687e2	8.687e2	-9.308e-1	-9.891e-1
	3.000	8.620e1	9.061e1	4.410	4.900e-3	9.050e1	8.713e2		8.620e1	8.724e2		-1.094	
Inoculum	1.000	7.873e1	1.060e2	2.726e1	2.550e-2	9.100e1	4.810e2		7.920e1	4.625e2		1.848e1	
	2.000	8.432e1	1.133e2	2.898e1	2.710e-2	9.887e1	5.364e2	5.364e2	8.474e1	5.208e2	5.208e2	1.559e1	1.848e1
	3.000	7.759e1	1.088e2	3.125e1	2.920e-2	9.344e1	5.420e2		7.818e1	5.216e2		2.038e1	

4.4.2 Density Determination

Density was determined by a gravimetric–volumetric method using graduated cylinders. Dry cylinders were weighed empty, then filled with distilled water at 20 °C and weighed again to verify the reading accuracy (density of water at $\rho_{20^{\circ}\text{C}} = 998.21 \text{ g L}^{-1}$). After drying, each cylinder was filled with a known volume of sample and weighed. The density of each material was calculated using

$$\rho = \frac{M_{\text{total}} - M_{\text{cylinder}}}{V_{\text{sample}}} \quad [\text{g L}^{-1}] \quad (4.4)$$

where M_{total} is the mass of cylinder plus sample, M_{cylinder} is the empty cylinder mass, and V_{sample} is the measured volume, the masses were measured in g and the volume in L. The results are on the table 4.2.

Table 4.2: Density values

Sample		M_{cylinder} (g)	M_{total} (g)	V_{sample} (L)	Density (g L^{-1})	Average Density (g L^{-1})
Food	1.000	1.360e2	2.420e2	9.920e−2	1.069e3	1.068e3
	2.000	1.160e2	2.230e2	1.002e−1	1.068e3	
Crude Glycerol	1.000	1.142e2	2.035e2	9.990e−2	8.932e2	8.931e2
	2.000	1.157e2	2.044e2	9.930e−2	8.929e2	
Inoculum	1.000	1.360e2	2.420e2	9.920e−2	1.069e3	1.068e3
	2.000	1.160e2	2.230e2	1.002e−1	1.068e3	

4.4.3 TOC and TN Determination

Total organic carbon (TOC), total carbon (TC) and total nitrogen (TN) were measured using a Shimadzu TOC-5000A analyser. Prior to analysis, the food waste was diluted 400 X and the crude glycerol was diluted 800 X to bring the concentrations within the linear range of the instrument. The results are on the table 4.3.

Table 4.3: Total carbon and Total nitrogen values

Sample		TOC (mg L^{-1})	TC (mg L^{-1})	Average TC (mg L^{-1})	TN (mg L^{-1})	Average TN (mg L^{-1})
Food	1.000	5.039	2.018e3	1.779e3	2.756e2	2.818e2
	2.000	3.927	1.540e3		2.879e2	
Crude Glycerol	1.000	8.808	7.050e3	5.523e3	4.700e2	4.886e2

Table 4.3 continued

Sample	TOC (mg L ⁻¹)	TC (mg L ⁻¹)	Average TC (mg L ⁻¹)	TN (mg L ⁻¹)	Average TN (mg L ⁻¹)
2.000	4.992	3.997e3		5.072e2	

4.5 Experiment Configuration

The reactor had a useful volume of 1 L and was operated as a continuous stirred-tank reactor (CSTR) under anaerobic conditions. Mixing was provided by a magnetic stir bar, and the overflow exit tube was kept closed with a valve to maintain an anaerobic environment.

The reactor was kept at 35 °C, close to the optimal temperature for mesophilic microorganisms (Choorit & Wisarnwan, 2007). The organic loading rate (OLR) was set with a max of 4 kg VS m⁻³ d⁻¹, with a hydraulic retention time (HRT) of 30 days, conditions for stable co-digestion of food waste and glycerol (Tomczak et al., 2025). The pH was maintained around 7 by incorporating KHCO₃ into the feed solution at a concentration of 10 g L⁻¹, and adding KHCO₃ to the reactor whenever the pH dropped lower than 6.

The organic loading rate (OLR) and hydraulic retention time (HRT):

$$\text{HRT} = \frac{V}{Q} \quad [\text{d}] \quad (4.5)$$

$$\text{OLR} = \frac{S_{\text{in}} \times Q}{V} \quad [\text{kg VS m}^{-3} \text{ d}^{-1}] \quad (4.6)$$

where S_{in} is the influent VS concentration (kg m⁻³), Q is the volumetric flow rate (m³ d⁻¹), and V is the reactor working volume (m³).

4.5.1 Feeding Solution

The feed was designed to achieve a carbon-to-nitrogen (C/N) ratio of 25 : 1, which is optimal for anaerobic digestion (Li et al., 2011). Characterisation of the food waste and crude glycerol showed that both substrates contained nitrogen, making it difficult to balance the C/N ratio using both of them together. Therefore, pure glycerol was used as an additive to allow precise control of the C/N ratio. Based on the chosen HRT and the daily volumetric flow rate, the reactor was fed once per day to ensure a flow rate high enough to transport the solid food-waste suspension through the feeding tube.

The feed was made with a restriction of needing to have less than 15 % of TS%. The amount of OLR and if KHCO₃ was added, was decided by the pH and how was the methane

production of the reactor. If it was too acidic, it was used a lower OLR with KHCO_3 , if it was at the right amount it was keep the same amount, and if it was too basic it was added a higher OLR without KHCO_3 . The feed storage always had some remainig feed, and the new one was added on top, diluting or concentrating the feed going to the reactor.

The feed was also suplemented with nutrients to complement the feed and create a better feed for the microorganisms. Table 4.4 shows the nutrients used and there ratios.

Table 4.4: Nutrients used and there ratios, for a 25:1 C/N. (Adapted from: Alqudah, 2025)

Nutrients	Mass (mg (350 mL) ⁻¹)	Mass (mg L ⁻¹)	Mass (g L ⁻¹)
MgSO4.7H2O	1.890e1	5.400e1	5.400e-2
FeSO4.7H2O	5.228e2	1.494e3	1.494
CoCl2.6H2O	4.200e-1	1.200	1.200e-3
ZnCl2	1.050e1	3.000e1	3.000e-2
NiCl2.6H2O	1.050e-2	3.000e-2	3.000e-5
EDTA	2.100e-1	6.000e-1	6.000e-4
CuSO4.5H2O	9.230	2.637e1	2.640e-2

4.5.2 Biogas Quantification

The *bioprocess control AMPTS II (Automatic Methane Potential Test System)* system (fig. 4.1), was provided to meassure the gas flow that was comming out of the reactor, with a CO_2 fixing unit using a solution of $3 \text{ mol dm}^{-3} \text{ NaOH}$.

4.5.3 Analysis Methods

It was used the method of DiLallo & Albertson (Ribas et al., 2007), to measure the alkalinity and volatile fatty acids. The sample collected was first centrifuged, at 2604 G for 30 min, to filter the particulated before doing the measurements.

$$\text{TA} = \frac{N_{\text{acid}} * V_{\text{acid}} * 50000}{V_{\text{sample}}} \quad [\text{mg CaCO}_3 \text{ L}^{-1}] \quad (4.7)$$

$$\text{VFA} = \frac{N_{\text{base}} * V_{\text{base}} * 60000}{V_{\text{sample}}} \quad [\text{mg CH}_3\text{COOH L}^{-1}] \quad (4.8)$$

The volumes were measured in mL, the sample volume was always 50 mL, the normality (N) of the acid and base, at first was 0.1, but then it was changed to 1. The eq. (4.7) is expressed in $\text{mg CaCO}_3 \text{ L}^{-1}$, and the eq. (4.8) is expressed in $\text{mg CH}_3\text{COOH L}^{-1}$.



(a) AMPTS gas collector



(b) AMPTS CO₂ trap



(c) AMPTS system

Figure 4.1: AMPTS II system used

4.5.4 Substrate used as feed

Table 4.5 shows all substrates used to feed the reactor and when they were made.

Table 4.5: Substrates used during the process

Day	Substrate 1	Substrate 2	Substrate 3	Extra Addition	COV & TS%	Volume Made
0 ^a	Food Waste, 156 g	Glucose, 1.9 g		Calcium Bicarbonate, 5 g	1.5 & 15	500 mL
4	Crude Glycerol, 23.2 mL	Pure Glycerol ^b , 31.5 mL		Potassium Bicarbonate, 1.9 g	0.8 & 10	250 mL
11	Food Waste, 12.54 g	Crude Glycerol, 9.02 mL	Pure Glycerol, 2.6 mL	Potassium Bicarbonate, 1.57 g	0.13 to 0.4 & 5 to 10 ^c	250 mL
18	Food Waste, 10.88 g	Crude Glycerol, 2.89 mL	Pure Glycerol, 1.44 mL		0.0256 & 2	250 mL
25	Food Waste, 31.12 g	Crude Glycerol, 44 mL	Pure Glycerol, 29 mL		1.4 & 10	250 mL
32	Food Waste, 31.01 g	Crude Glycerol, 44 mL	Pure Glycerol, 28.9 mL		1.4 & 10	250 mL
39	Food Waste, 46.52 g	Crude Glycerol, 66 mL	Pure Glycerol, 43 mL	Potassium Bicarbonate, 2.6 g	3.1 & 15	250 mL
46	Food Waste, 50.07 g	Crude Glycerol, 66 mL	Pure Glycerol, 43 mL	Potassium Bicarbonate, 2.67 g	3.16 & 15	250 mL
53 ^d	Crude Glycerol, 55 mL	Pure Glycerol, 11.5 mL		Potassium Bicarbonate, 5.38 g	0.42 & 10	500 mL
60	Crude Glycerol, 55 mL	Pure Glycerol, 12 mL		Potassium Bicarbonate, 5.18 g	0.42 & 10	500 mL ^e
67	Crude Glycerol, 41 mL	Pure Glycerol, 9 mL		Potassium Bicarbonate, 2.56 g	0.95 & 15	250 mL

Table 4.5 continued

Day	Substrate 1	Substrate 2	Substrate 3	Extra Addition	COV & TS%	Volume Made
74	Crude Glycerol, 60 mL	Pure Glycerol, 12.5 mL		Potassium Bicarbonate, 2.57 g	1.37 & 15	250 mL

^a It only started feeding on day 2.

^b The solution of pure glycerol was diluted 100 X.

^c The food waste wasn't well mixtured this day.

^d Change of substrate.

^e There was a leak on the feeding tube.

Chapter 5

Results

5.1 Reactor Condition and Visual Observations

The reactor was inoculated with anaerobic sludge and initially contained a murky, light brown suspension (fig. 5.1a). Within three days of operation, the colour changed to black (fig. 5.1b), which is characteristic of active anaerobic digestion and is typically caused by the precipitation of metal sulphides, particularly iron sulphide, under reducing conditions. This visual change indicated the establishment of a suitable anaerobic environment and the onset of methanogenic activity.

At the end of the experimental period, after approximately two weeks of unattended operation, four discrete masses resembling fungal colonies were observed on the internal surfaces of the reactor during cleaning (figs. 5.2a and 5.2b). The origin of these growths is uncertain, but they may have been introduced via the food waste feed or during periods when the feeding tube was opened for repairs (e.g. leaks) and when changing the feed that the tube had to be cleaned. The presence of such organisms could have competed with the anaerobic microbial community for substrate, potentially affecting biogas production in the later stages of the experiment.

5.2 Biogas Production

Biogas production was monitored continuously using AMPTS II, which recorded both the cumulative volume of methane produced and the instantaneous flow rate. The entire experimental period is summarised in figs. 5.3 and 5.4. To facilitate interpretation, the data have been divided into four distinct phases based on the feeding regime: (i) an adaptation phase 5.2.2, (ii) co-digestion of food waste and glycerol 5.2.3, (iii) digestion of crude glycerol 5.2.4, and (iv) a post-feeding period 5.2.5.



(a) Murky suspension



(b) Black suspension

Figure 5.1: Visual changes in the reactor.



(a) Masses inside the dry reactor



(b) Masses floating in the reactor

Figure 5.2: Visual for the masses inside the reactor.

5.2.1 Overall Production

The cumulative methane production over the whole experiment (fig. 5.3) shows a continuous increase, indicating that the reactor maintained methanogenic activity throughout the monitored period. The total flow profile (fig. 5.4) exhibits distinct peaks corresponding to daily feeding events, with variations in amplitude reflecting the different substrate compositions and organic loading rates applied during each phase. And a outlier peak at day 53, that was when the feed was changed from a mixture of food waste and crude glycerol, to only crude glycerol, and due to some complications, there was that anomaly.

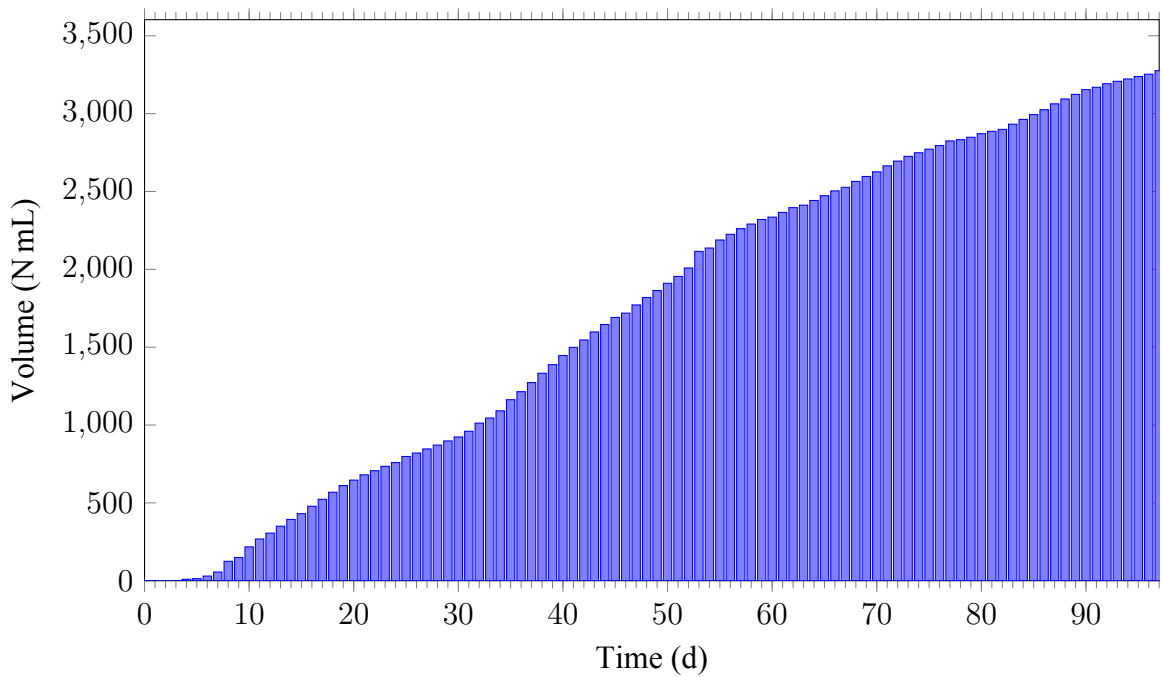


Figure 5.3: Cumulative volume of CH₄ N mL

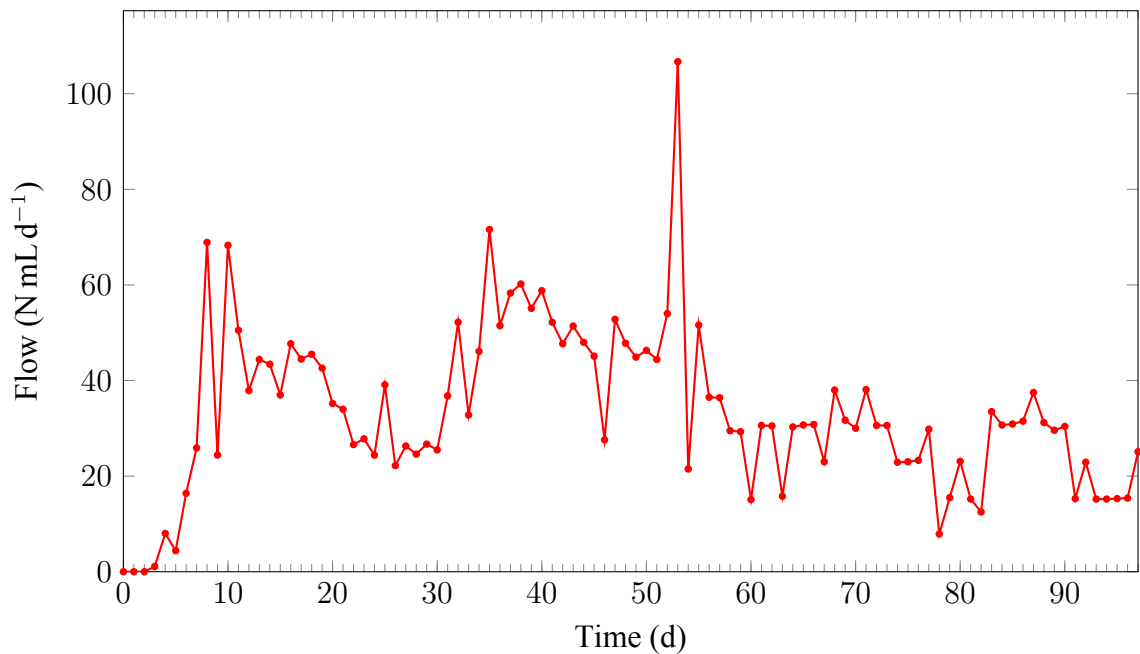


Figure 5.4: Volume flow of CH₄ N mL d⁻¹

5.2.2 Adaptation Phase

During the initial adaptation phase, biogas production was low (figs. 5.5 and 5.6). This lag period is normal for anaerobic reactors as the microbial community adjusts to the new substrate and operating conditions. The small but measurable methane flow confirmed that the viability of the inoculum, and the gradual increase in flow suggested the onset of exponential growth of the microorganisms.

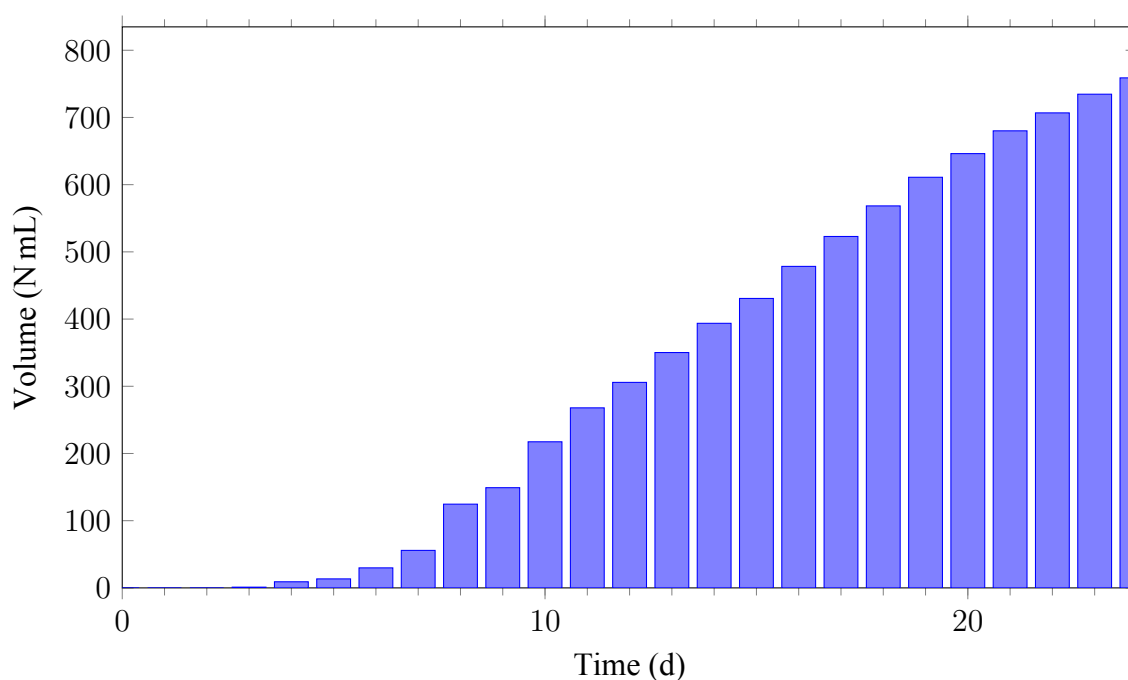


Figure 5.5: Cumulative volume of CH₄ N mL (Adaptation Phase)

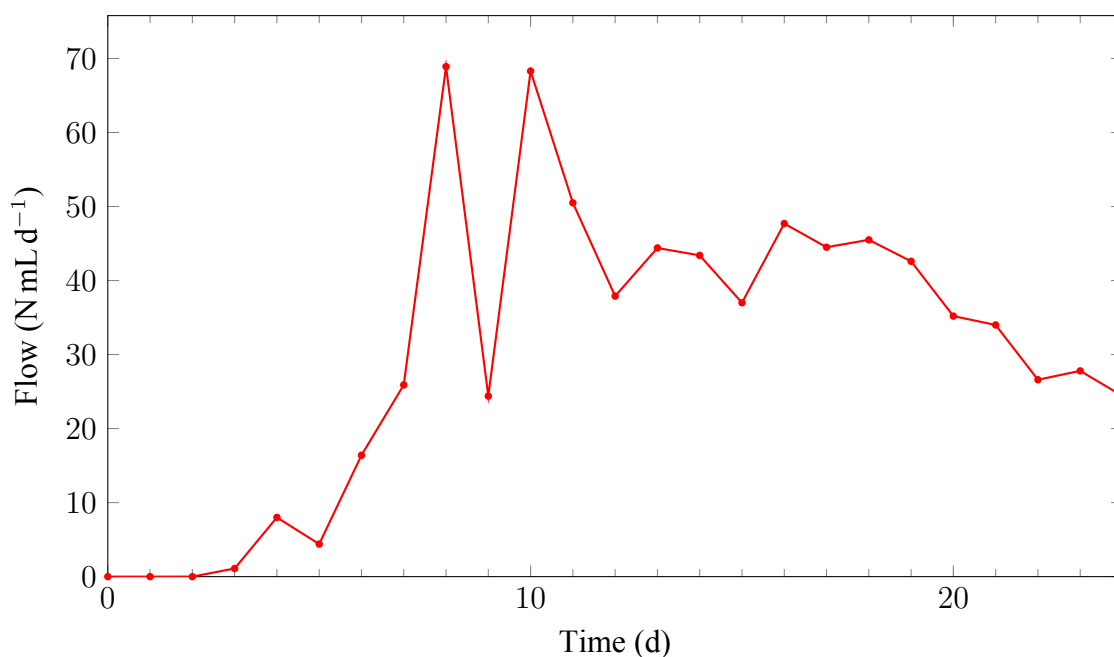


Figure 5.6: Volume flow of CH₄ N mL d⁻¹ (Adaptation Phase)

5.2.3 Co-digestion of Food Waste and Crude Glycerol

The reactor was fed with a mixture of food waste and crude glycerol with a higher concentration, a marked increase in biogas production was observed (figs. 5.7 and 5.8). The cumulative methane volume rose steeply, and the flow rate showed pronounced spikes immediately after each feeding event (fig. 5.9). These spikes are typical of readily degradable substrates, which are quickly hydrolysed and converted to volatile fatty acids and subsequently to methane. The average flow rate during this phase was higher than in any other phase, indicating that the co-digestion of food waste and glycerol provided a balanced nutrient supply and avoided the inhibition often associated with high glycerol concentrations alone.

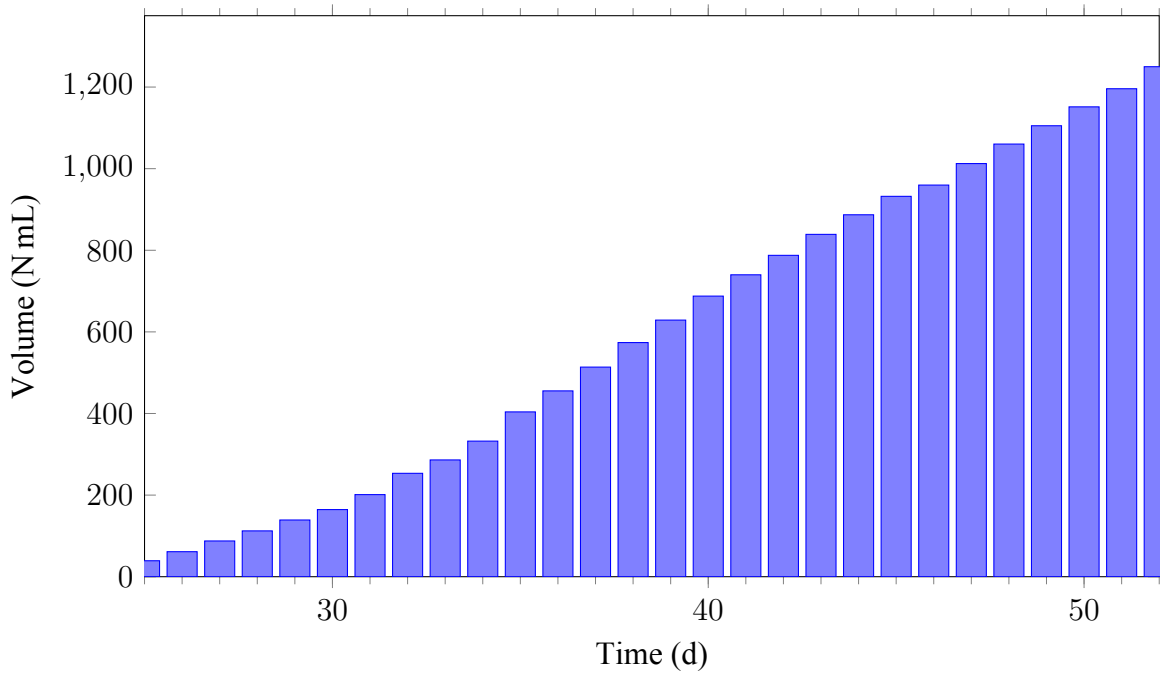


Figure 5.7: Cumulative volume of CH₄ N mL (co-digestion of food waste and crude glycerol)

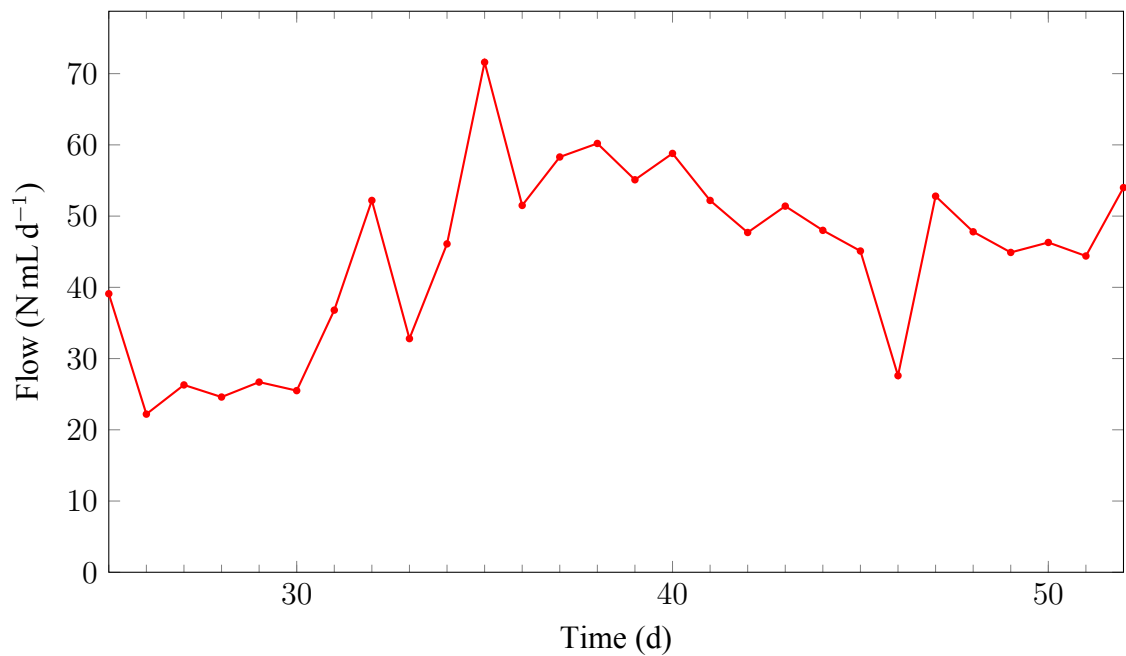


Figure 5.8: Volume flow of CH₄ N mL d⁻¹ (co-digestion of food waste and crude glycerol)

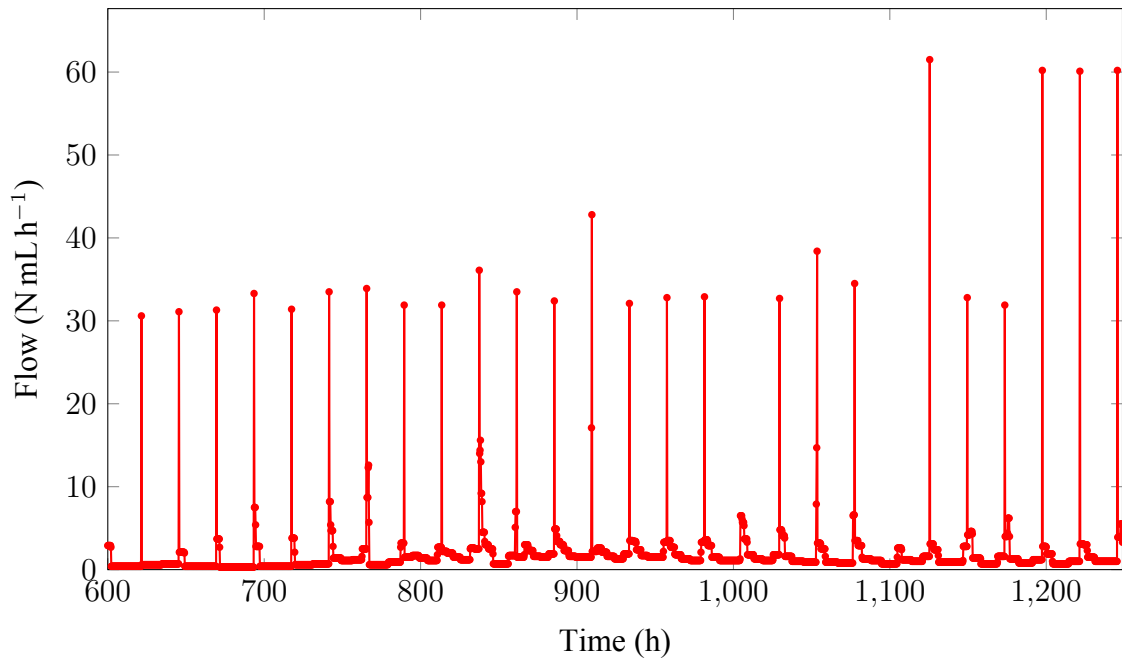


Figure 5.9: Volume flow of CH_4 N mL h^{-1} (co-digestion of food waste and crude glycerol)

5.2.4 Digestion of Crude Glycerol

When the feed was changed to crude glycerol alone, biogas production decreased noticeably (figs. 5.10 and 5.11). Although spikes in flow were still observed after feeding (fig. 5.12), their magnitude was bigger than during the co-digestion phase, and the cumulative methane volume plateaued more quickly. Several factors may explain this lower performance:

- The glycerol could have been transformed too fast or too slow, to maintaining a concentration of intermediate products.
- The retention rate of $\frac{1}{30} \text{ d}^{-1}$ may have been insufficient to wash out inhibitory by-products or to maintain a stable population of glycerol-fermenting bacteria.
- The appearance of fungal masses towards the end of this phase could have further stressed the anaerobic community by competing for substrate or introducing toxic metabolites.

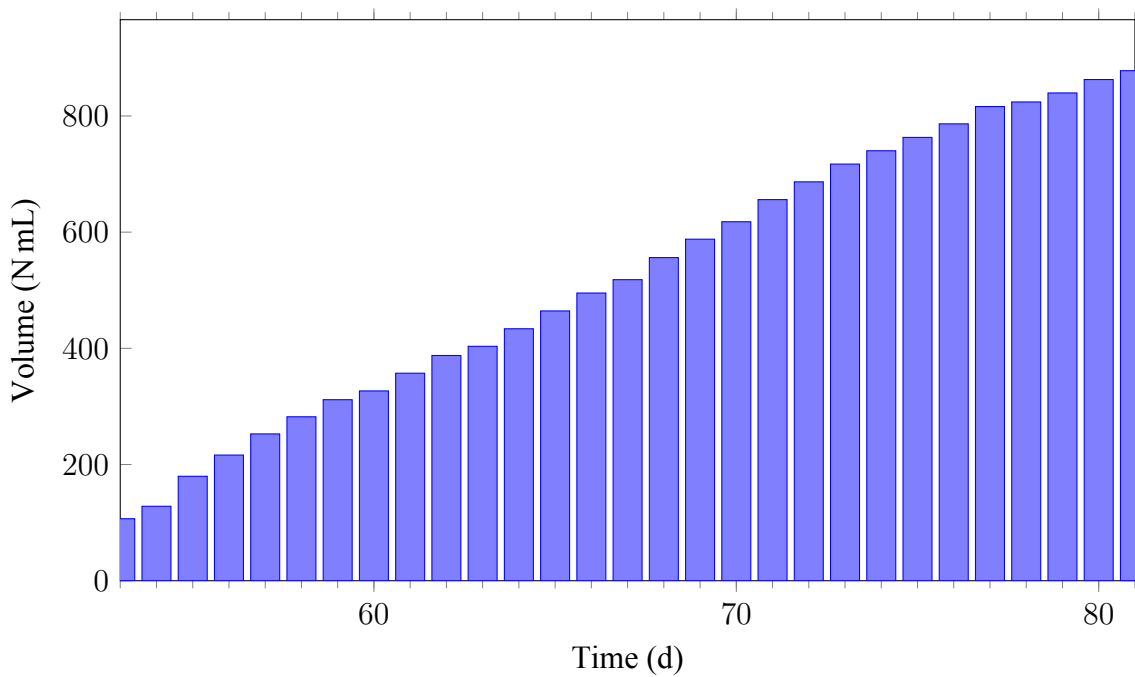


Figure 5.10: Cumulative volume of CH_4 N mL (Anaerobic digestion of crude glycerol)

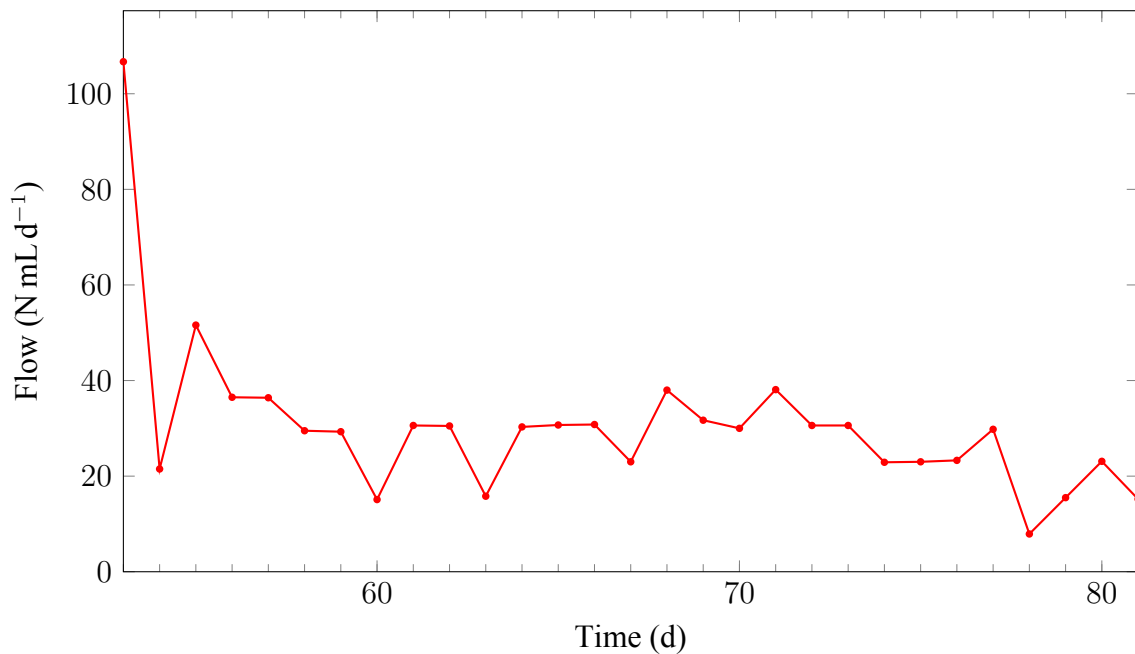


Figure 5.11: Volume flow of CH_4 N mL d^{-1} (Anaerobic digestion of crude glycerol)

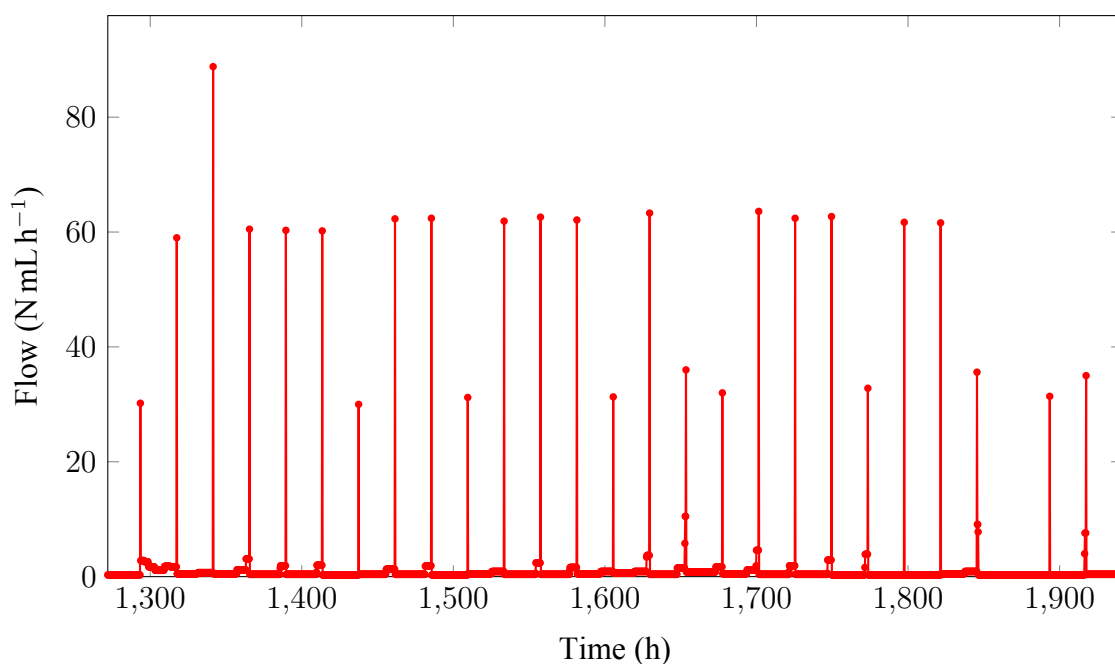


Figure 5.12: Volume flow of CH₄ N mL h⁻¹ (Anaerobic digestion of crude glycerol)

5.2.5 Post-feeding Period

During the final two weeks, no fresh substrate was added to the reactor, it just had some residual substrate from the glycerol only phase for a week. Nevertheless, a small residual biogas production was recorded (figs. 5.13 and 5.14). This production can be attributed to the slow degradation of accumulated organic matter and microbial decay products. The flow rate declined gradually, confirming that the reactor was no longer being actively fed and that the remaining substrate was limited.

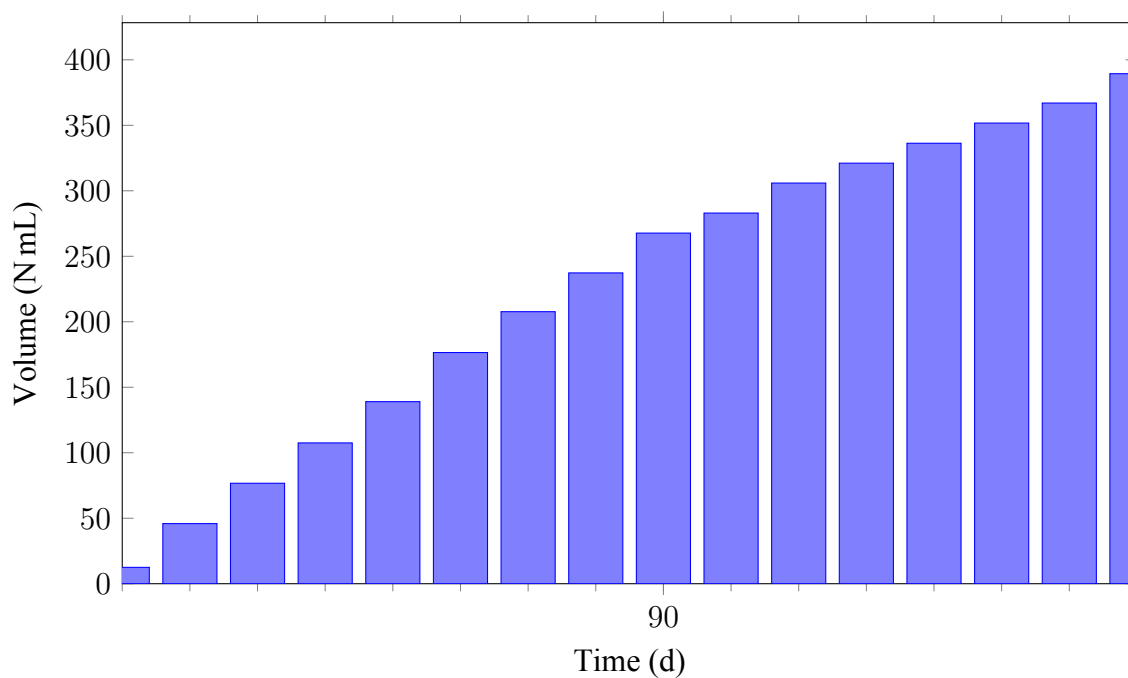


Figure 5.13: Cumulative volume of CH₄ N mL (Post-feeding period)

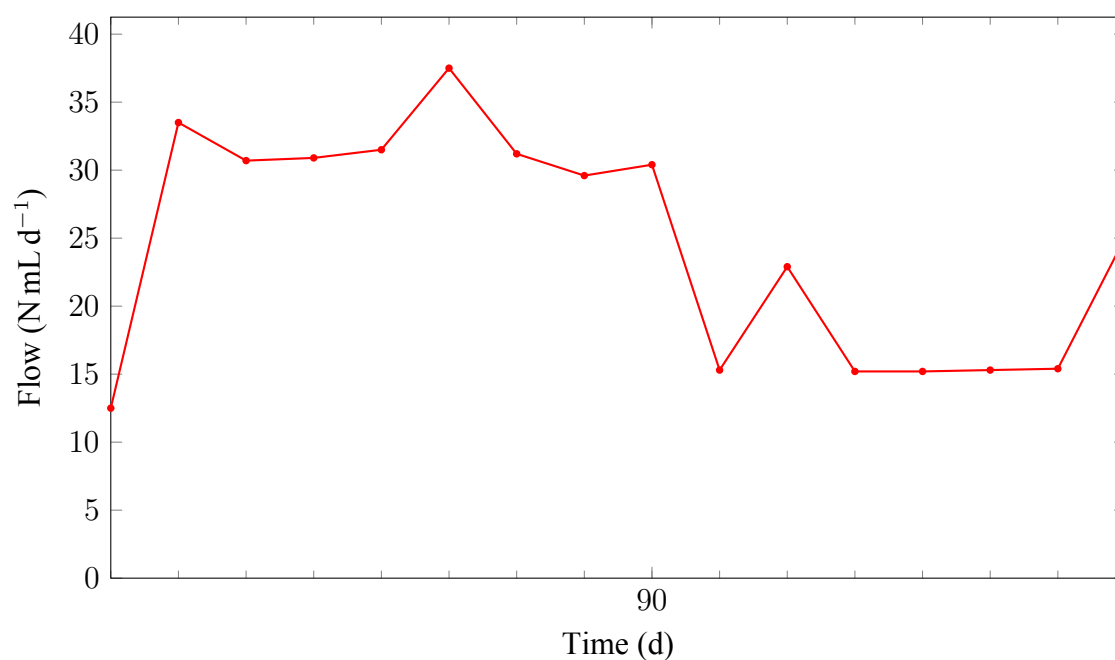


Figure 5.14: Volume flow of CH₄ N mL d⁻¹ (Post-feeding period)

5.2.6 Alkalinity and Volatile Fatty Acids

The total alkalinity and VFA concentrations were monitored throughout the experiment to assess process stability. The results are shown in fig. 5.15. Initially, both alkalinity and VFA

were low, reflecting the early adaptation phase before buffer addition. After the manual introduction of KHCO_3 , alkalinity increased sharply to values above $7000 \text{ mg CaCO}_3 \text{ L}^{-1}$ and remained relatively high until close to the end. This high alkalinity was maintained by the addition of bicarbonate to the feed, which helped to counteract the acidity produced during the degradation of the feed.

VFA concentrations also increased after the start of feeding, reaching values between 3000 to $5000 \text{ mg CH}_3\text{COOH L}^{-1}$. Such high VFA levels are often associated with process imbalance, as they can inhibit methanogenesis. However, the high alkalinity provided a strong buffering capacity, and the pH remained above 6.5 for most of the experiment, with the co-digestion the pH was falling and with glycerol only the pH was increasing. The *VFA/alkalinity* ratio is a more reliable indicator of instability. In this experiment, the ratio fluctuated between 0.3 to 1.2 . While using the co-digestion feed a ration around 0.8 to 1.0 had the highest production.

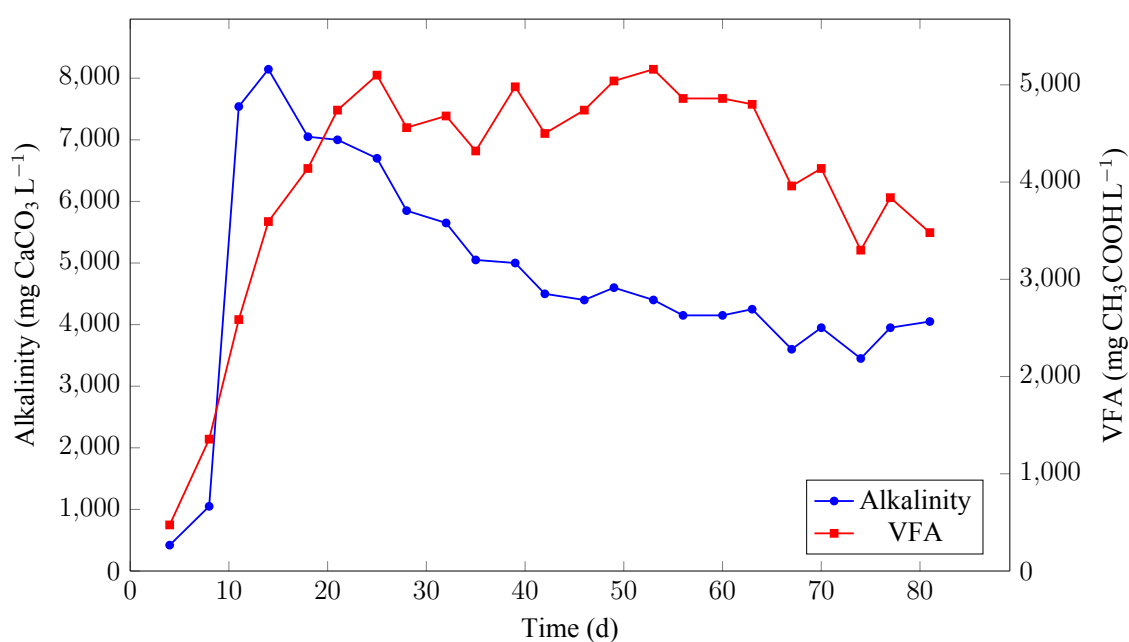


Figure 5.15: Source: Author

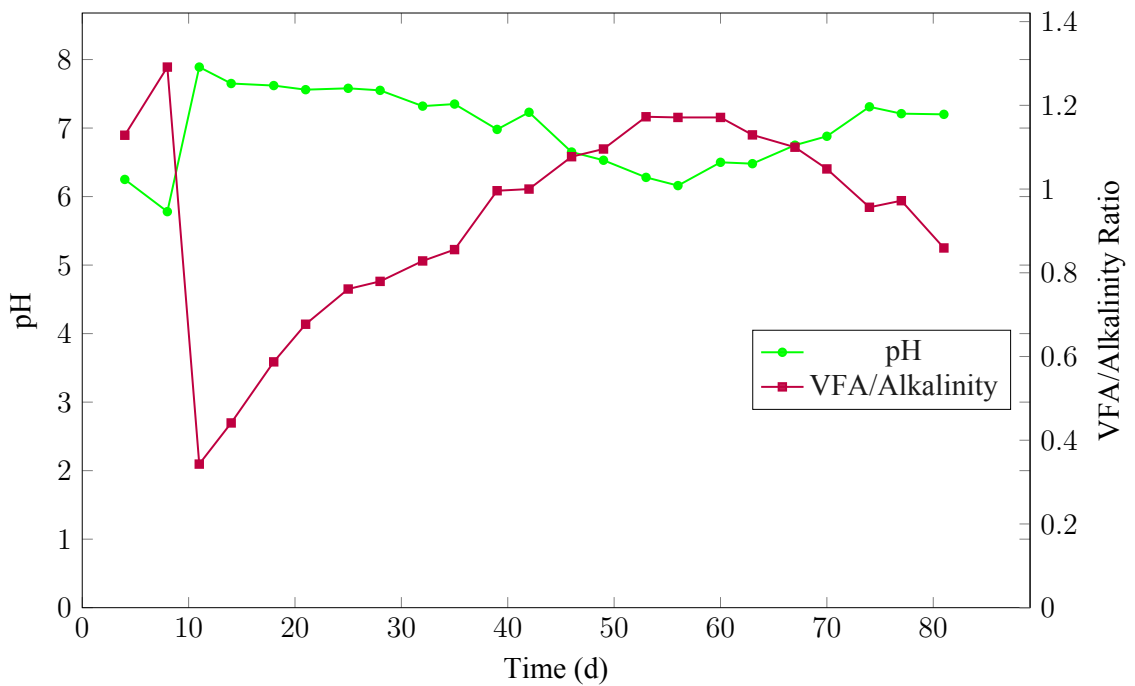


Figure 5.16: Source: Author

Chapter 6

Conclusion

A direct comparison of the co-digestion and glycerol only phases reveals that the co-digestion of food waste and glycerol resulted in higher biogas yields and more stable operation. The pH was decreasing faster with the co-digestion, where with glycerol only phase it wasn't decreasing that fast, and was even growing with the potassium bicarbonate that was added to maintain the pH. These results support the hypothesis that glycerol, when co-digested with food waste, can enhance biogas production by balancing the C/N ratio and providing readily available carbon, whereas glycerol alone may work, but would need a different setup and adjustments.

Using crude glycerol, with some pure glycerol to make the necessary adjustment to balance the C/N ratio, could be a viable solution for the glycerol problem. It's only major problem, is that crude glycerol contains a lot of impurities that can affect the anaerobic digestion, and for a system that can reliably consume it, it will need a better control of it's substrate.

Future studies could involve on how to use better the crude glycerol as a feed, to try subside it's downside of it's impurities and need for an extra carbon rich substrate to balance the feed.

Abbreviations

Symbols

Units

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